Intrusion Detection Systems Using Machine Learning

Project Report

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Abstract

An Intrusion Detection System (IDS) is a critical component of modern network security, designed to monitor and analyze network traffic for signs of unauthorized access or malicious activity. This study evaluates various machine learning models for intrusion detection using the UNSW-NB15 data set, a comprehensive and challenging benchmark comprising over 2.5 million records and featuring nine distinct attack categories. The methodology involves data preprocessing, including feature selection, label encoding, scaling, and addressing class imbalance with the Synthetic Minority Oversampling Technique (SMOTE). Multiple models such as Logistic Regression, Random Forest, Naive Bayes variants, Decision Tree, Bagging, Boosting (XGBoost), Stacking, and Voting Ensembles are assessed for their detection performance. The analysis reveals that tree-based ensemble methods (Random Forest and Decision Tree) achieved the highest accuracy and F1 scores, demonstrating strong performance in attack identification.

Keywords: Intrusion Detection System (IDS), Machine Learning, Network Security, UNSW-NB15 data set, Feature Selection, Label Encoding, Scaling, Class Imbalance, SMOTE, Attack Identification.

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Introduction

An Intrusion Detection System (IDS) is a vital component of modern network security architectures, designed to continuously monitor and analyze network traffic to identify signs of unauthorized access, malicious activity, or policy violations. As cyber threats become increasingly sophisticated and pervasive, traditional security measures such as firewalls and antivirus software are no longer sufficient to ensure robust protection. IDSs serve as an additional layer of defense, capable of detecting threats that may bypass primary security mechanisms and providing early warnings to administrators or security teams so that swift defensive actions can be taken.

A key challenge in IDS deployment is the detection of a wide variety of attack types, including those that exploit unknown vulnerabilities or use sophisticated evasion techniques. To address this, research in the field often utilizes comprehensive datasets, such as the UNSW-NB15 data set, which contains millions of records and a diverse array of attack categories. Such datasets enable the evaluation and improvement of IDS algorithms, particularly in handling issues like class imbalance and the detection of rare or emerging attack types.

It is an indispensable tool in the cybersecurity landscape, providing organizations with the ability to detect, alert, and respond to threats that might otherwise go unnoticed. As cyber attacks continue to evolve, IDS technologies are also advancing, integrating machine learning, real-time analytics, and adaptive detection methods to keep pace with the dynamic threat environment.

Modern IDS technologies are increasingly incorporating advanced machine learning and artificial intelligence techniques to enhance their detection capabilities. Their deployment is not only critical for protecting sensitive data and infrastructure but also for meeting regulatory compliance requirements in sectors such as finance, healthcare, and critical infrastructure. As cyber attacks continue to evolve, IDS technologies are advancing in tandem, integrating real-time analytics, adaptive detection methods, and seamless interoperability with other security solutions to keep pace with the dynamic threat environment.

Related Work

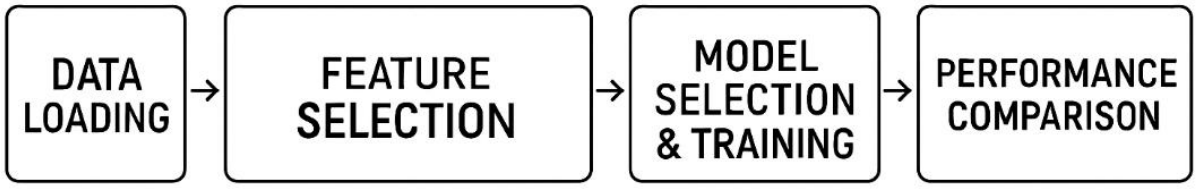
Recent research on intrusion detection systems (IDS) using the UNSW-NB15 data set highlights the effectiveness of machine learning and ensemble methods for detecting a wide range of network attacks. The UNSW-NB15 dataset has become a cornerstone for modern intrusion detection research.

Thakkar and Lohiya [1] introduced an ensemble-learning-based deep neural network for classifying imbalanced intrusion data in IoT networks. Most research leverages structured datasets to classify attack types or identify anomalies in real-time. Moustafa and Slay [2] introduced the UNSW-NB15 data set, which has become a widely adopted benchmark for evaluating intrusion detection systems due to its diversity of modern attack categories and realistic traffic patterns. Their follow-up analysis [3] compared UNSW-NB15 with the older KDD99 data set and demonstrated that the latter lacked up-to-date attack scenarios and suffered from redundancy issues. Sarhan et al. [5] focused on NetFlow-based datasets tailored for machine learning-based intrusion detection, highlighting the importance of scalable data and flow-level features for real-time detection systems. In addition, More et al. [6] analyzed the performance of multiple IDS models on the UNSW-NB15 data set, revealing that ensemble classifiers especially when combined with oversampling techniques like SMOTE significantly improve detection of rare and minority attack classes. Feature engineering innovations significantly boosted performance, with Hammad et al. demonstrating clustering-classification hybrids for optimal feature selection [7], and Rosy-Kumar achieving 95.2% accuracy using Validated Feature Selection [10]. Advanced architectures like Sun et al.'s PSO-optimized CNN-BiLSTM-BiGRU [8] and More et al.'s stacked ensembles [5] pushed detection accuracy beyond 98%, confirming tree based ensemble superiority over traditional models.

Methodology

Overview

This study presents a comprehensive methodology for developing and evaluating an Intrusion Detection System (IDS) using the UNSW-NB15 data set, with a focus on addressing class imbalance and comparing multiple machine learning models for network attack detection.



Data Acquisition and Preparation

The UNSW-NB15 data set, a modern and challenging benchmark for IDS research, was used. It contains 2,540,044 records with 49 features, split into training (175,341 records) and testing (82,332 records) sets.

The dataset includes nine attack categories (e.g., Fuzzers, DoS, Exploits) and a binary label indicating attack or normal traffic. The nine categories of attacks are as follows-

1. Fuzzers
Fuzzers involve sending malformed or random data to a system to crash or disrupt it. According to the dataset's distribution, Fuzzers attacks account for approximately 14.7% of all attack records, making them one of the more prevalent attack types represented. Specifically, the dataset contains 24,246 Fuzzers records out of over two million total entries.

2. Analysis
Analysis attacks involve analyzing network traffic or system vulnerabilities to identify potential entry points. It represents a group of threats that typically target web applications and network services through various means such as

open ports, emails, and scripts.In the UNSW-NB15 dataset, Analysis attacks are specifically labeled and make up about 1.6% of all attack records.

3. Backdoors
Backdoors involve creating a hidden entry point in a system to gain unauthorized access. Backdoor attacks are relatively rare in the dataset, comprising about 1.4% of all attack records.

4. DoS (Denial of Service)
DoS attacks involve overwhelming a system with traffic to make it unavailable. DoS attacks account for approximately 9.9% of all attack records in the data set, making them a significant category for intrusion detection research.

5. Exploits
Exploits involve exploiting known vulnerabilities in software or systems. According to the dataset's attack category distribution, Exploits account for approximately 27% of all attack records.

6. Generic
Generic attacks refer to a broad class of network attacks designed to exploit cryptographic weaknesses or vulnerabilities in protocols, often without targeting a specific application or service. Generic attacks often involve exploiting cryptographic protocols or mechanisms, such as brute-force attacks on encryption, hash collisions, or other weaknesses that are not tied to a specific service or software vulnerability. The Generic category is the most common attack type in the UNSW-NB15 data set, making up approximately 35.8% of all attack records.

7. Reconnaissance
Reconnaissance attacks involve gathering information about a system or network to identify potential vulnerabilities. Reconnaissance attacks account for approximately 8.5% of all attack records in the data set, making them a significant, though not dominant, class.

8. Shellcode
Shellcode attacks involve injecting malicious code into a system to gain control. Reconnaissance attacks account for approximately 8.5% of all attack records in the data set, making them a significant, though not dominant, class.

9. Worms
Worms involve self-replicating malware that spreads from system to

system.Worm attacks are extremely rare in this data set, making up only about 0.1% of all attack records.

These attack types are used to evaluate the performance of intrusion detection systems and other cybersecurity tool

Data preprocessing

Data Loading-

Load the training and test datasets from the UNSW_NB15 data files. The data set is split into a training set (UNSW_NB15-training) and a test set (UNSW_NB15-testing).

Data Cleaning-

Removing Redundant and Duplicate Records:
In the data preprocessing phase for intrusion detection using the UNSW-NB15 data set, one of the initial and crucial steps is to check for and remove redundant or duplicate records. This process ensures that the data set is free from repeated entries, which could otherwise introduce bias and distort the training of machine learning models. By eliminating these duplicates, the data quality is improved, and the resulting models are more reliable and generalizable.
Handling Missing Values:
Missing data can reduce model accuracy, create misleading visualizations, and introduce bias into the model. Hence, missing values are imputed, commonly using strategies like mean imputation, to maintain data set integrity.

Data Transformation-

Encoding Categorical Features:
The UNSW-NB15 data set contains several categorical features—such as protocol type (proto), service (service), state (state), and attack category (attack_cat)—that represent qualitative information. Machine learning algorithms typically require numerical input, so these categorical variables must be transformed into numeric representations before model training. One-Hot Encoding is used where each category is converted into a separate binary feature (column) that takes the value 1 if the record belongs to that category, and 0 otherwise.

Converting all values to numeric:
After encoding, all columns are converted to numeric data types to ensure compatibility with scikit-learn and other ML libraries.

Feature Selection-

Exclude non-numeric and identifier columns such as:
id, proto, service, state, attack_cat, and label (for binary classification tasks).

For multi-class tasks, encode categorical variables (like proto, service, state, attack_cat) using one-hot encoding.

Scaling-

Feature scaling is a crucial preprocessing step in machine learning that transforms the range of independent variables so all features contribute equally to model training. Hence, feature scaling is applied to standardize the range of independent variables.

Handling Class Imbalance-

SMOTE (Synthetic Minority Oversampling Technique) is used to address class imbalance. SMOTE generates synthetic samples for minority classes, balancing the class distribution and improving model performance on underrepresented classes. SMOTE generates synthetic samples for the minority class by interpolating between existing minority class instances, rather than simply duplicating them. For each minority class sample, SMOTE identifies its k-nearest minority neighbors in feature space. It then randomly selects one neighbor and creates a new synthetic sample at a point along the line segment connecting the original sample and its neighbour. This process is repeated until the desired class balance is achieved.

Model Training and Evaluation

The following models were trained and compared:

Single Classifiers:

- Logistic Regression: Logistic regression is a supervised classification algorithm that predicts binary outcomes by modeling the probability of an event. It transforms linear combinations of input features into probabilities between 0 and 1, using a decision boundary.

- Random Forest: A tree-based ensemble method that constructs multiple decision trees during training.

- Bernoulli Naive Bayes: Optimized for binary features (0/1), calculating class probabilities based on feature presence/absence.

- Gaussian Naive Bayes: Assumes continuous features follow a normal distribution, estimating probabilities using mean and variance.

- Decision Tree: A flowchart-like structure where nodes represent features, branches denote decision rules, and leaves indicate outcomes.

Ensemble Methods:

- Bagging: Trains multiple base models (e.g., decision trees) on random data subsets sampled with replacement.

- XGBoost (Boosting): A sequential boosting algorithm where each model corrects errors from prior models.

- Stacking: Combines diverse base models (e.g., SVM, logistic regression) by training a meta-model on their predictions.

- Voting: Aggregates predictions from heterogeneous models using:
  1.Hard voting: Majority class wins.
  2.Soft voting: Averages class probabilities.

Results

Performance was quantified using:

Accuracy: Proportion of correctly classified instances.
The formula for accuracy is:
Accuracy = (TP + TN) / (TP + TN + FP + FN) (…1)
Where:
- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Precision: True positives among predicted positives (minimizes false alarms).
The formula for precision is:
Precision = TP / (TP + FP) (…2)
Where:
- TP = True Positives
- FP = False Positives

Recall (Sensitivity): True positives detected among actual positives (critical for threat detection).
The formula for recall is:
Recall = TP / (TP + FN) (…3)
Where:
- TP = True Positives
- FN = False Negatives

F1-Score: Harmonic mean of precision and recall, ideal for imbalanced data.
The formula for F1 score is:
F1 = 2 × (Precision × Recall) / (Precision + Recall) (…4)
Where:
- Precision = TP / (TP + FP)
- Recall = TP / (TP + FN)

Performance Comparison

I. Performance Comparison Table (Binary Detection)

Classifier	Accuracy	Precision	Recall	F1 Score
Logistic Reg.	0.78	0.73	0.95	0.82
Random Forest	0.92	0.89	0.96	0.93
Bernoulli NB	0.75	0.85	0.66	0.74
Gaussian NB	0.73	0.70	0.91	0.79
Decision Tree	0.92	0.91	0.95	0.93
Bagging (DT)	0.84	0.78	0.99	0.88
XGBoost	0.87	0.82	0.99	0.90
Stacking Ens.	0.88	0.84	0.98	0.90
Voting Ens.	0.89	0.84	0.98	0.90

Ensemble and tree-based models such as Random Forest, Decision Tree, XGBoost, Stacking, and Voting consistently outperformed simpler models like Logistic Regression and Naive Bayes on the UNSW-NB15 data set, achieving higher accuracy and F1 scores. Random Forest reached an accuracy of 0.9168 and F1 score of 0.9273, while Decision Tree and XGBoost also delivered strong results above 0.92 and 0.89, respectively.

Observation

The comparative analysis focuses on the performance of various machine learning models for intrusion detection using the UNSW-NB15 data set. The data set is imbalanced, with some attack types (e.g., Worms, Shellcode) being extremely rare, which necessitated the use of SMOTE for oversampling minority classes such as Analysis, Backdoor, Shellcode, Worms while the majority classes are Generic, Exploits, Fuzzers, DoS, Reconnaissance.

The observation drawn from the UNSW-NB15 binary classification experiments and supporting literature highlight several key insights about model performance, data characteristics, and practical implications for network intrusion detection. Across all tested algorithms, ensemble and tree-based methods such as Random Forest, Decision Tree, XGBoost, Stacking, and Voting consistently outperformed simpler linear and probabilistic models like Logistic Regression and Naive Bayes. tree-based and ensemble models. Decision Tree, Random Forest, Voting, Stacking, XGBoost, and Bagging provided the best balance between detecting attacks and minimizing false alarms. These models are recommended for practical intrusion detection tasks due to their superior ability to differentiate between attack and non-attack events.

A major factor influencing these results is the inherent class imbalance in the UNSW-NB15 dataset, where attack records are much less frequent than normal traffic. The use of SMOTE and other resampling techniques was critical for balancing the dataset, which in turn significantly improved recall and F1 scores for minority (attack) classes. However, while these techniques boost the detection of rare attacks, they can sometimes lead to reduced precision due to the introduction of synthetic samples and increased class overlap. This trade-off was evident in models like Bagging with Decision Trees, which achieved a high recall of 0.9935 but a lower precision of 0.7829, indicating a tendency to over-predict the attack class.

Conclusion

This project successfully implemented and evaluated multiple machine learning models for binary classification of network traffic as attack or non-attack using the UNSW-NB15 data set. Among all models tested, tree-based and ensemble approaches such as Decision Tree, Random Forest, Voting, and Stacking classifiers demonstrated the highest effectiveness, achieving accuracies above 88% and F1 scores above 0.90 in several cases. These models also maintained strong recall values, indicating a high capability to detect attack instances. In contrast, traditional models like Logistic Regression and Naive Bayes, while achieving reasonable recall, showed lower precision and overall accuracy. The results highlight that ensemble and tree-based methods are better suited for intrusion detection tasks on this data set, offering a robust balance between detecting attacks and minimizing false positives. Overall, the findings provide a clear direction for deploying effective machine learning solutions in network security environments.

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